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Palaeomagnetic results from the Tibetan Sedimentary Series of the Manang area (north central Nepal)

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SUMMARY

The remanent magnetization of Palaeozoic and Mesozoic carbonates from the Tibetan Sedimentary Series (TSS) of the Manang area north of the Annapurna Range (north central Nepal) has been investigated. The TSS forms part of the tectonic unit between the Indian–Eurasian plate boundary in the north and the Main Central Thrust (MCT) in the south.

Apart from a present field direction, three remanence components have been separated: a normal polarity, post-folding remanence (negative fold test) is carried by pyrrhotite (the core corrected mean direction for five sites with 47 specimens from Silurian/Devonian rocks is decl. = 312.9°, incl. = 58.3°, $\alpha_{95} = 17.6°$, $k = 19.8$). The origin of this component is not understood.

A more precisely determined, reversed polarity, post-folding remanence (99 per cent negative fold test) is also carried by pyrrhotite (the core corrected mean direction for 12 sites with 229 specimens from Carboniferous/Triassic rocks is decl. = 196.4°, incl. = −65.9°, $\alpha_{95} = 3.2°$, $k = 183$).

One site of Lower Carnian age shows—in addition to the pyrrhotite component—a second, stable remanence direction carried by magnetite. The fold test is 99 per cent positive and there is a high probability that the remanence is of detrital origin (the bedding corrected direction for one site with 40 specimens is decl. = 334.3°, incl. = −54.1°, $\alpha_{95} = 4.2°$, $k = 30.2$, palaeolatitude 34.6°S).

The direction of the magnetite component coincides with other Triassic palaeodirection data from the literature and the inclination suggests a palaeoposition of the TSS at the northern margin of India at this time. The northern extent of India in the Triassic cannot be reconstructed from the data because of the strong anticlockwise rotation (48°) of stable India since this time.

The inclination of the reverse polarity pyrrhotite component is 19° steeper than expected from the maximum northern position of the area. This suggests a regional northward tilt along the MCT which is supported by steep inclinations found in other areas.

Declination data for the reverse polarity pyrrhotite component and for the magnetite component deviate clockwise from expected values for the area by 23°–34° (calculated from the Indian APWP). Utilizing Klootwijk, Conaghan & Powell's (1985) rotational underthrusting model, the minimum magnitude of continental underthrusting between the TSS and stable India at the longitude of central Nepal (84°E) has been calculated from the declination of our reverse polarity pyrrhotite component. A possible range of 520–1100 km results from consideration of different tectonic models.

Key words: Himalaya, northern India margin development, palaeomagnetism, Tibetan Sedimentary Series.

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INTRODUCTION

The India–Eurasia convergence, associated with subduction of oceanic crust at the Eurasian margin, is well documented by the apparent polar wander paths for both continents (Irving & Irving 1982; Klootwijk 1984). In particular the rapid northward drift of India during the Maastrichtian to Eocene (15–20 cm yr^{-1}) is well established from sea floor spreading data from the Indian Ocean, suggesting a collision age of about 50 Ma (Patriat & Achache 1984). Fig. 1 shows the principal tectonic units in the Himalayan region. An ophiolite zone along the Indus–Yarlung Tsangpo is generally accepted as representing the boundary between the Eurasian and Indian plates (the Indus Tsangpo Suture Zone: ITS). Near the end of the Oligocene the zone of subduction propagated to the south, to take the form of intercontinental underthrusting at the Main Central Thrust (MCT, the main underthrusting zone) and possibly later to the Main Boundary Fault (e.g. Le fort 1975; Fuchs 1982). Palaeozoic to Eocene sediments of the Tibetan Sedimentary Series (TSS) between the ITS and the MCT are generally interpreted as shelf sediments deposited on the passive northern margin of the Indian plate. A quantitative determination of the amount of crustal shortening between the TSS and stable cratonic India requires the reconstruction of the former northern margin of India, prior to collision. Only few palaeomagnetic data have been published from the TSS (Klootwijk & Bingham 1980; Klootwijk *et al.* 1983; Besse *et al.* 1984; Zhu Zhiwen & Teng Jiwen 1984). The data are subject to many uncertainties, resulting from mechanical deformation and metamorphic overprinting of the sediments and much more work is required to establish a reliable data base. In this paper we report palaeomagnetic investigations from the TSS of the Manang area, north of the Annapurna Range (north central Nepal; Fig. 1).

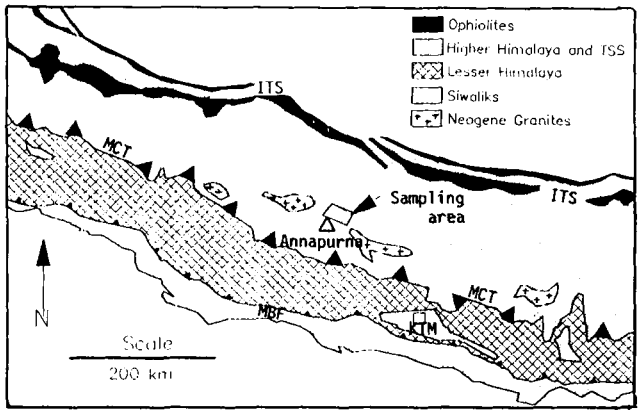


Figure 1. Simplified tectonic map of the Central Himalaya (from Gansser 1980) showing the sampling area of Manang, north of the Annapurna Range. ITS: Indus Tsangpo Suture Zone, MCT: Main Central Thrust, MBF: Main Boundary Fault, KTM: Kathmandu.

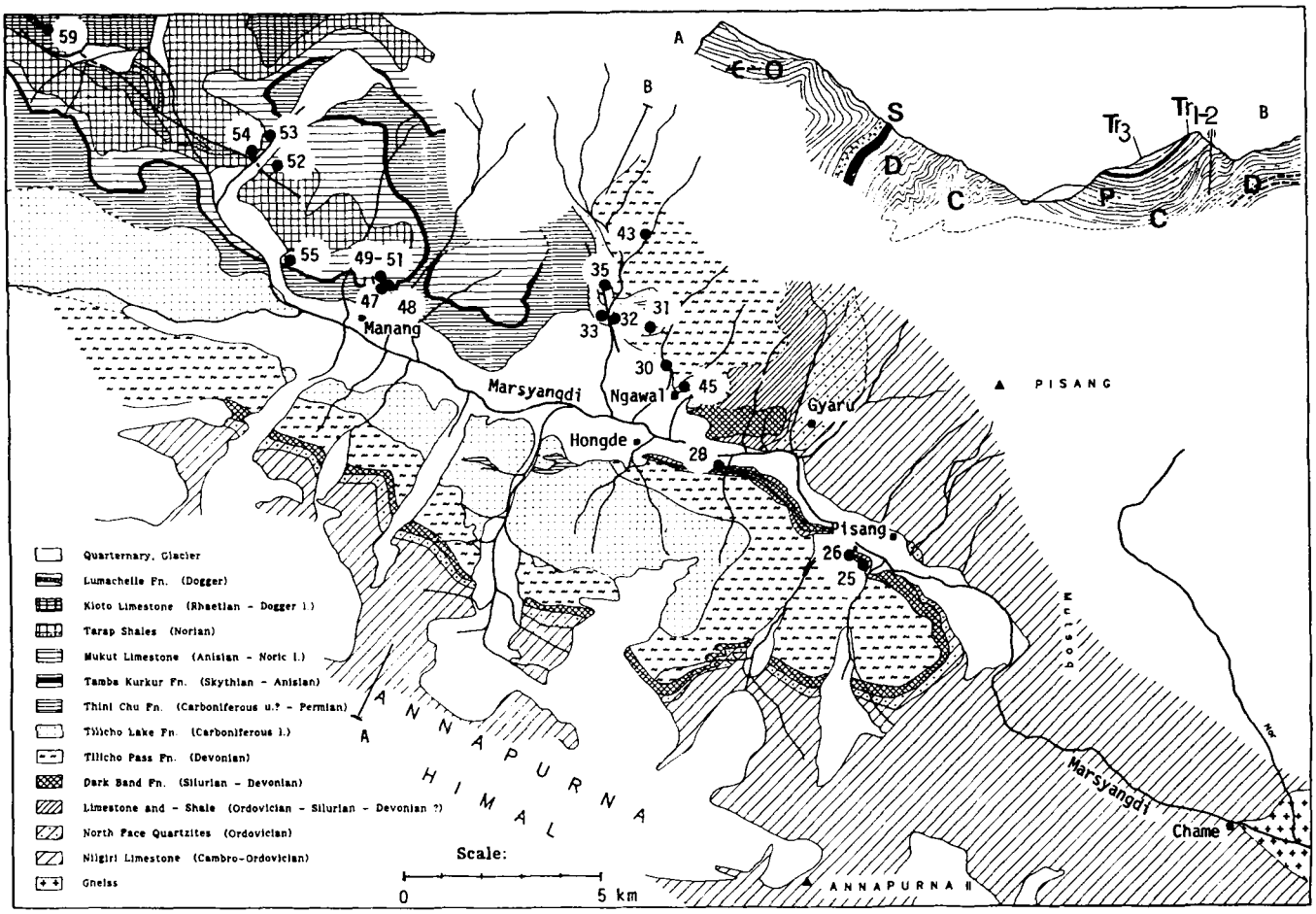


Figure 2. Simplified geologic map of the Manang area and SSW–NNE cross-section A–B (after Fuchs *et al.* 1988) showing the locations of sites discussed in this paper (numbered black dots).

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LOCAL GEOLOGY AND SAMPLING

Sampling was carried out in April 1987 during a joint project of palaeomagnetic, geologic (mapping, stratigraphy) and petrographic (metamorphism, structural analyses) investigations. The geologic work revealed a fold and thrust geometry much simpler than the one previously published by Bordet, Colchen & Le Fort (1975). It consists of large syncline along the Marsyangdi valley with an WNW-dipping fold axis (Fuchs, Widder & Tuladhar 1988; Fig. 2). In this paper we refer to the model of Fuchs *et al.* because their work incorporates the French contribution and also additional tectonic and biostratigraphic results.

The stratigraphic sequence encloses Cambro-Ordovician (mainly in the east) to Jurassic (in the west) sediments (Figs 2 and 3). Petrographic investigations (Schneider & Masch 1990 and personal communication) showed—contrary to the suggestion of Bordet *et al.* (1975)—that metamorphism does not fade away in the Silurian–Devonian rocks. Metamorphic temperatures of about 500 °C were established for the Lower Palaeozoic rocks, about 400 °C in the Carboniferous rocks, and greenschist metamorphism has even affected the Mesozoic carbonates (temperatures are deduced from calcite–dolomite thermometry, illite crystallinity and metamorphic minerals).

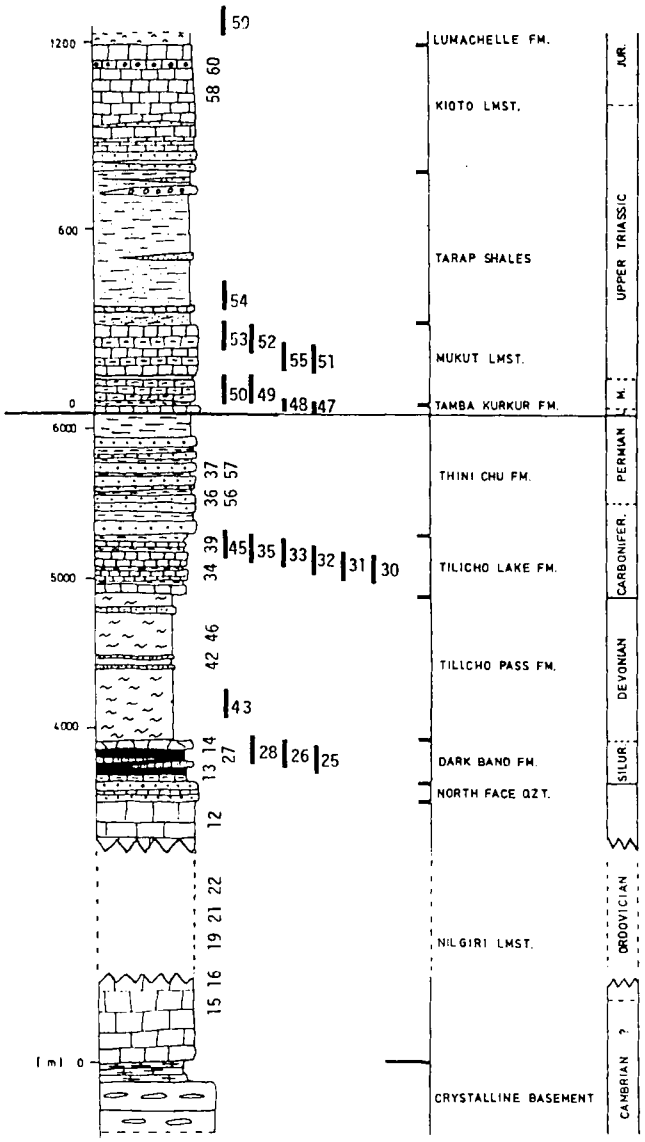


Figure 3. Stratigraphic sequence of the Tibetan Sedimentary Series of the Manang area [after Bordet *et al.* (1975) and Fuchs *et al.* (1988)]. Stratigraphic positions of sites discussed in this paper are shown with bars and horizontal numbers. Sites from which no results could be obtained are written vertically.

Sampling was carried out throughout the entire stratigraphic sequence (Fig. 3) using a portable drill. Orienting of the core samples was done with a magnetic compass (the generally low magnetization intensities did not require solar bearings). Dating of the site locations is based on Bordet *et al.* (1975) and on biostratigraphic analyses (mainly of conodonts) by the third author (see also Fuchs *et al.* 1988).

No results could be obtained from Ordovician and most of the Silurian, Devonian, Permian and Jurassic rocks. Either demagnetization shows an unstable magnetization, remanence directions within sites are highly scattered, or only a present field component is recorded. The stratigraphic positions of these sites are shown in Fig. 3, but details are not discussed in this paper.

Reliable and interpretable remanence directions have been obtained from 12 sites (Carboniferous sites 30, 31, 32, 33, 35, Triassic sites 47, 49, 50, 51, 52, 53, 55). An additional eight sites show results with a lower reliability or with an uncertain interpretation. Site localities, stratigraphic positions, lithologies and bedding data are listed in Table 1 and show in Figs 2 and 3.

GENERAL SAMPLE TREATMENT

Cylindrical cores were sliced to standard dimensions (diameter 2.5 cm, length 2.2 cm). From each site two pilot specimens per core sample were progressively demagnetized by thermal and alternating field methods. In case of an uncertain demagnetization behaviour another pair of specimens was studied. Susceptibility measurements followed each step of thermal treatment to identify mineral alterations.

Depending on the results of the pilot studies the remaining specimens were demagnetized in 5 to 10 suitable steps. Generally one specimen from each core sample was measured. Orthogonal vector projections, principal component analyses, equal area projections and Fisher statistics were used for the determination of remanence components and for the calculation of mean directions.

For thermal demagnetization a μ-metal shielded SCHONSTEDT furnace was used. Alternating field demagnetization was carried out with a HIGHMOOR tumbling demagnetizer using a Helmholtz coil system for field compensation.

Measurements of the remanence directions were generally carried out with a cryogenic magnetometer (Rémanomètre à squid RS 01, LETI; noise level about 10^{-2} mA m^{-1}).

MAGNETIC MINERALS

Rock magnetic experiments have been carried out to identify the magnetic minerals: isothermal remanence

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Table 1. Geological ages, locations, bedding data (direction of dip arrows, degrees; f: sampled from a fold structure), geological units and lithologies of sites discussed in this paper.

SITE	AGE	BEDDING strike dip	LOCATION	FORMATION	LITHOLOGY
25	S	10-210 37-85 f	1 km SW Pisang	Dark Band Fn.	pure gray limestones, partially siliceous
26	S	180-210 40-57	1 km SW Pisang	Dark Band Fn.	calcareous, siliceous schists; dark limestones
28	S	212-270 25-36	2.5 km SW Gyaru	Dark Band Fn.	limestones, siliceous limestones and calcareous quartzites
43	D	224-241 30-36	4 km NNW Ngawal	Tilicho Pass Fn.	siliceous limestones, calcareous quartzites, quartzites;
30	Cl	212-248 20-43	0.75 km NNW Ngawal	Tilicho Lake Fn.	dark limestones and calc-schists
31	Cl	235-294 10-20	1.5 km NNW Ngawal	Tilicho Lake Fn.	dark limestones with argillaceous interlayers
32	Cl	336-354 32-44	2.5 km NW Ngawal	Tilicho Lake Fn.	dark limestones with argillaceous interlayers
33	Cl	255-290 7-20	2.7 km NW Ngawal	Tilicho Lake Fn.	dark limestones and marly-schists
35	Cl	206-228 28-36	3 km NW Ngawal	Tilicho Lake Fn.	light-gray and dark limestones and calc-schists
45	Cl	224-344 16-36	0.1 km NE Ngawal	Tilicho Lake Fn.	dark bluish limestones with light-gray argillaceous schists
47	Tr_1m	15-354 2-7	1 km NE Manang	Tamba Kurkur Fn.	light-gray limestones
48	Ansl	320-21 2-6	1 km NE Manang	Tamba Kurkur Fn.	light-gray limestones
49	Ansl/Ansm	74-278 2-14	1 km NE Manang	Mukut Limestone	dark argillaceous limestones
50	Ansu	18-38 6-30	1 km NE Manang	Mukut Limestone	dark-gray argillaceous limestones
51	Crnl	74-338 3-20	1 km NE Manang	Mukut Limestone	dark-gray argillaceous limestones
55	Crnl	14-24 55-85 f:limb$_1$ 234-308 10-20 f:limb$_2$	2.5 km WNW Manang	Mukut Limestone	dark limestones and argillaceous limestones
52	Crnm/Crnu	150-264 5-20	4.3 km NW Manang	Mukut Limestone	dark bluish limestones and marls
53	Crnm/Crnu	190-240 26-35	5 km NW Manang	Mukut Limestone	dark bluish limestones and marls
54	Nor	0 0	4.8 km NW Manang	Tarap Shales	dark silt- and sandstones
59	J_2	205-217 38-51	3 km WNW Churi Lattar	Lumachelle Fn.	dark calc-schists, dark-gray argillaceous limestones

(IRM) acquisition and thermal demagnetization of a saturation IRM (SIRM) and of the natural remanence (NRM), low-temperature measurements of IRM, scanning electron microscopy with energy dispersive analyses (EDA), thermomagnetic measurements (J_s–T curve) with Curie temperature determination, and domain structure observations.

Pyrrhotite was found to be the dominating ferrimagnetic mineral, identified by the following.

(i) IRM acquisition curves with saturation fields of about 0.6–0.8 T (Fig. 4B: sites 26, 35, 55).

(ii) Rapid decrease of NRM (Fig. 8) and SIRM (Fig. 4A) in the temperature ranges 300–350 °C and 300–400 °C

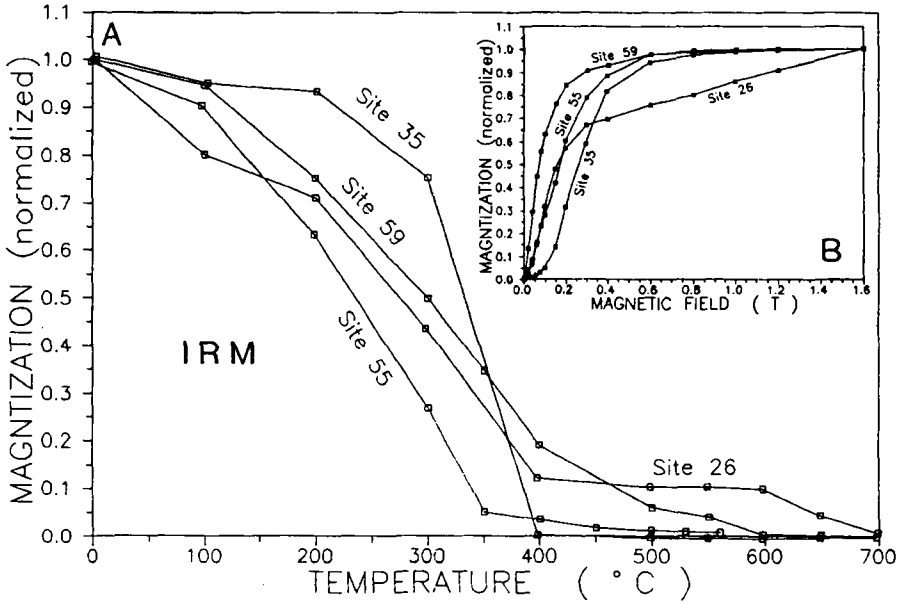


Figure 4. Isothermal remanence acquisition (B) and thermal demagnetization of saturation IRM (A) for representative specimens (site 26: Silurian, site 35: Carboniferous, site 55: Lower Carnian, site 59: Dogger).

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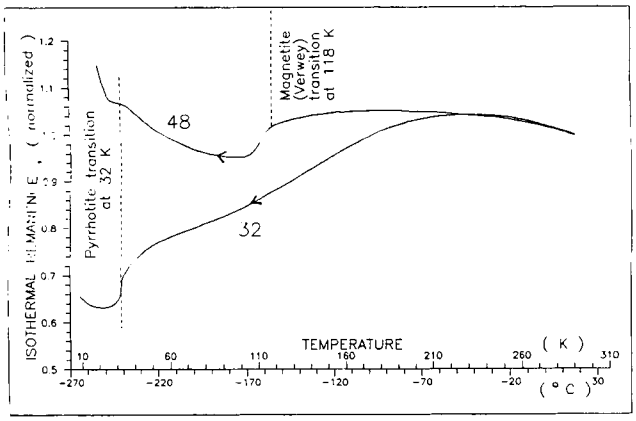


Figure 5. Isothermal remanence of two samples at low temperatures. Transitions at 32 K (pyrrhotite) and 118 K (magnetite) are evident for site 32 (Carboniferous) and site 48 (Lower Anisian).

respectively. This temperature range corresponds approximately to the Curie temperature of pyrrhotite (325 °C). The occasional occurrence of slightly higher blocking temperatures has been observed in earlier studies and may be interpreted as representing pyrrhotite–magnetite interactions (Linssen 1988; P. Rochette, personal communication).

(iii) Low-temperature transition of IRM at 32 K (Fig. 5) which is typical for pyrrhotite (Rochette 1988).

(iv) J_s–T curve of non-separated material from site 43 (Fig. 6) shows the Curie temperature of pyrrhotite (however, pyrrhotite could be not identified in J_s–T curves of any separated fractions!).

(v) EDA of measured ore particles show peaks of Fe and S.

(vi) Domain structure observations show domain patterns (Fig. 7) typical of pyrrhotite (Soffel 1981).

Magnetite occurs besides pyrrhotite, identified by the following.

(i) Residual magnetization (NRM or SIRM) after thermal demagnetization of pyrrhotite. The residual remanence disappears at about the Curie temperature of magnetite (578 °C) (Fig. 4: sites 55, 59; Fig. 8: site 55).

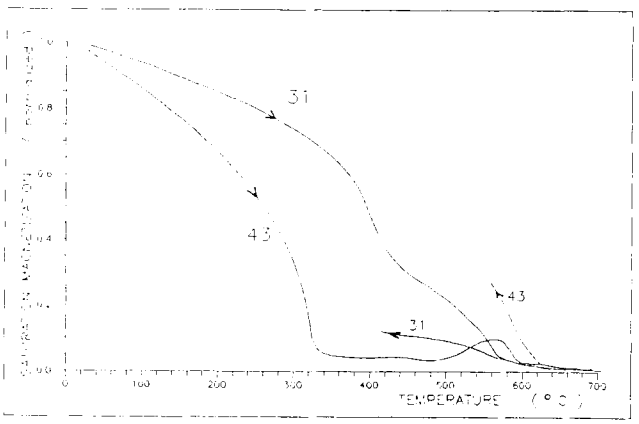


Figure 6. Temperature dependence of the saturation magnetization (J_s–T curve) showing the Curie temperature of pyrrhotite (site 43) and magnetite (site 31). The decrease at about 400 °C can be referred to maghemite decay.

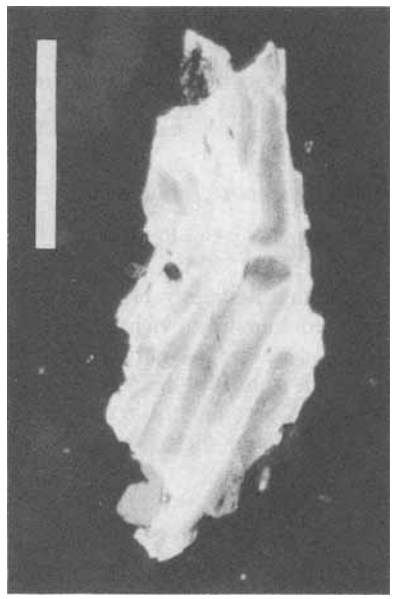


Figure 7. Domain structures typical of pyrrhotite are shown by the Bitter pattern technique. Scale bar: 50 μm.

(ii) Lower coercive component in IRM acquisition curves (Fig. 4: site 59, 55?).

(iii) Low-temperature (Verwey) transition of SIRM at 118 K (Fig. 5).

(iv) J_s–T curves of separated material show the Curie temperature of magnetite (Fig. 6: site 31). The decrease of J_s at about 400 °C probably results from maghemite decay and indicates the presence of low-temperature oxidized magnetite.

(v) EDA of measured ore particles show a peak of Fe only.

Haematite occurs in some rock units, identified by high saturation fields if IRM (>1.6 T) and a drop of SIRM between 650 and 700 °C (Fig. 4: site 26).

The rock magnetic studies allow a definite identification of the ferrimagnetic minerals within the samples. All Palaeozoic and most of the Triassic sites (except 49, 54) contain mainly pyrrhotite. Some haematite is found in the Silurian rocks and possibly in the Devonian site 43. A minor contribution of magnetite appears in the Carboniferous carbonates. The magnetite fraction increases in Triassic rocks, culminating in sites 49, 54 and especially in the Jurassic site 59.

The volume ratio of pyrrhotite to magnetite increases in rocks of greater geological age. This is probably a function of an increasing metamorphic grade. In the view of Rochette (1988), pyrrhotite is formed from magnetite or from pyrite in low metamorphic grade sediments. Carboniferous rocks have been affected by higher metamorphism temperatures than Mesozoic formations and more magnetite has been converted.

Hence conservation of a primary detrital remanence is expected only in cases where detrital magnetite is sufficiently preserved and its remanence is not affected by thermal or mechanical overprinting. Pyrrhotite is probably of secondary origin and hence will carry a secondary remanence.

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REMANENCE ANALYSIS

As well as a present field component, three principal groups of remanence directions have been separated.

Post-folding reverse pyrrhotite component

The majority of the sites (Carboniferous: 30, 31, 32, 33, 35; Triassic: 47, 48, 50, 51, 52, 53, 55) show a demagnetization behaviour similar to that demonstrated by site 31 in Fig. 8. All sites contain predominantly pyrrhotite. It is likely that pyrrhotite was formed during the period when metamorphism was active with temperatures of about 400 °C in Carboniferous rocks. Consequently a thermoremanence is suggested for the pyrrhotite direction after cooling below the blocking temperature.

Thermal demagnetization proved to be generally suitable for separating principal remanence components. Viscous remanences or present field components—if existing—are demagnetized at 200 °C. Up to about 350 °C the remanence directions and intensities are highly stable (remanence predominantly carried by pyrrhotite). The residual magnetization (magnetite) after demagnetization of the pyrrhotite component is usually very weak or unstable.

Pyrrhotite directions within individual specimens have been determined by averaging the directions within the range of stable intensity between 200 °C and the blocking temperature of pyrrhotite (if necessary by difference vectors). 97 per cent of all measured specimens are included in the Fisher statistics. The excluded specimens show in part NRM intensities near the measurement limit or differ strongly from the site mean directions for unknown reasons.

The resulting site mean directions are listed in Table 2. Site 55 shows a negative intrasite fold test at the 99 per cent confidence level (McElhinny 1964). A 99 per cent negative fold test is also evident for the site mean directions (Fig. 9). Hence a post-folding age of remanence acquisition is proved. The mean core corrected (cc) direction of the 12 sites shows a reverse polarity with steep (negative) inclination (reverse pyrrhotite component). The cc directions show lower precisions of mean directions within individual sites ($k = 21.6$–179, mean: 72) compared to the site mean directions ($k = 183$). This suggests that secular variations were recorded within the sites.

Magnetite-rich sites (Triassic: 49, 54 Jurassic: 59) show also a steeply inclined reversed polarity pyrrhotite component, analysed from thermal (site 54) or alternating field (sites 49, 59) demagnetization results. However, these data are not included in the Fisher statistics of the reverse pyrrhotite component because of weak remanence intensities or because of a different demagnetization behaviour in part of the samples (Fig. 8: site 59).

Post-folding normal pyrrhotite component

Site 45 (Carboniferous), site 43 (Devonian) and sites 25, 26, 28 (Silurian) contain predominantly pyrrhotite with some haematite. The demagnetization behaviour is similar to that of the sites discussed above. However a normal polarity pyrrhotite component is recorded (Fig. 8: site 25). Above the blocking temperature of pyrrhotite the residual magnetization carried by haematite becomes generally unstable.

Remanence directions within individual specimens have been determined in the same way as for the reverse pyrrhotite component. 87 per cent of all specimens are included in the Fisher statistics (following the same criteria for accepting specimens as for the reverse pyrrhotite component).

The resulting site mean directions are listed in Table 2 and plotted in Fig. 9. Site 25 shows a negative intrasite fold test. The mean directions also tend to a negative fold test, but the confidence level is below 95 per cent. The age of remanence acquisition certainly post-dates folding.

The overall mean direction of five sites shows a normal polarity, also with steep (positive) inclination (normal pyrrhotite component). However it is far less precise ($k = 19.8$) than the reverse pyrrhotite component. The mean directions of the normal and reverse pyrrhotite components are not antiparallel; declinations differ by 63.5°. A different age or mechanism of remanence acquisition of unknown nature must be responsible.

Pre-folding magnetite component

In about 40 per cent of the specimens from the Lower Carnian site 55 a second stable component is broken down between the blocking temperature of pyrrhotite (Fig. 8) and the Curie temperature of magnetite (magnetite component). Site 55 was studied extensively (128 specimens). Magnetite remanence directions of individual specimens have been determined by averaging directions in the range 400–500 °C (above 500 °C the susceptibility increases).

How can suitable and non-suitable specimens be distinguished? In Fig. 10 the intensity of the magnetite component (at 400 °C: RM400) is plotted versus the NRM intensity (corresponding to the pyrrhotite plus the magnetite component). The specimens are classified into a group with a stable magnetite component (category 1) and others, with an unstable magnetite component or with RM400 near or below the measuring limit (category 2). This preliminary classification is somewhat subjective for several specimens. However NRM intensitites and ratios of RM400/NRM are about 3–4 times higher for category 1 than for category 2. Variations of the mineralogical composition (e.g. clay content) within the site may account for it. Metamorphism has probably affected magnetite particles differently in rock parts with different mineralogical compositions. The final decision to accept or reject specimens as representing a stable magnetite component has been made by setting lower limits for NRM ($0.5\,\mathrm{mA\,m^{-1}}$) and RM400 ($0.1\,\mathrm{mA\,m^{-1}}$). This reduces the number of suitable specimens from 52 to 40, but improves the objectivity of the selection.

The magnetite component passes a positive fold test on the 99 per cent level (Table 2, Fig. 11) and hence pre-dates folding. The bedding corrected (bc) mean direction is well defined ($k = 30.2$).

Anisotropy of magnetic susceptibility χ(AMS) has been measured on samples from site 55 to detect possible correlations between the mean remanence directions and the fabric of the rock. Values of χ are within a constant range independent of the intensity of NRM (Fig. 12A). Hence χ probably arises from paramagnetic minerals rather

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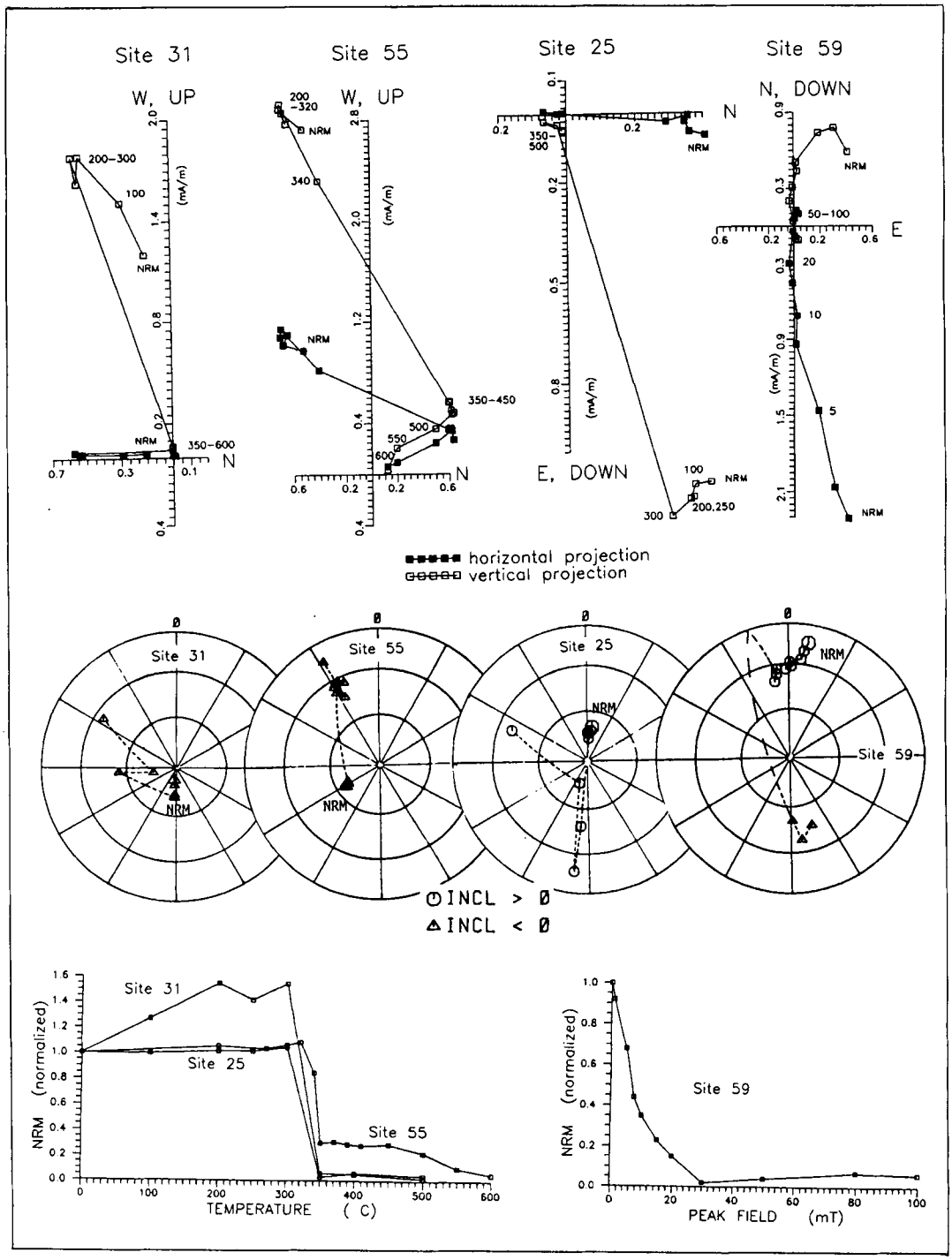


Figure 8. Thermal (site 31: Carboniferous, site 55: Lower Carnian, site 25: Silurian) and alternating field (site 59: Dogger) demagnetization results of representative specimens, demonstrated by orthogonal vector projections (top), equal area projections (middle) and remanence intensities (bottom). Directions before bedding correction are shown. Demagnetization steps are given in (°C) or (mT).

than ferrimagnetic particles. The temperature dependence of χ (Fig. 12B) shows a nearly linear curve of $1/\chi(T)$ which also suggests a paramagnetic control of χ. However the slight deviation of the $1/\chi(T)$ curve near room temperature suggests that ferrimagnetic minerals also contribute to χ.

The anisotropy degree $P = \chi_{max}/\chi_{min}$ is quite low (1.032 ± 0.024). Scattering of the maximum and minimum χ-axes is highly pronounced (Fig. 13). However a largely sedimentary type fabric is indicated by the trend of a vertical orientation of the minimum χ-axes and flat lying maximum χ-axes of the bedding corrected susceptibility tensor.

In summary, the rock fabric, as far as it is indicated by

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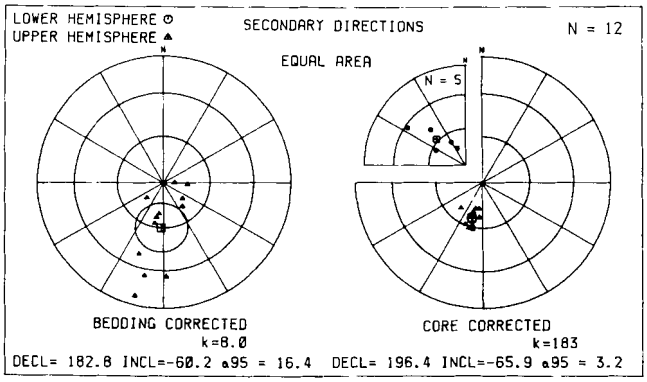


Figure 9. Equal area projection of the reverse pyrrhotite component. Mean directions of 12 sites before and after bedding correction are shown. A 99 per cent negative fold test is evident. The small projection shows mean core correction directions of the normal pyrrhotite component (five sites).

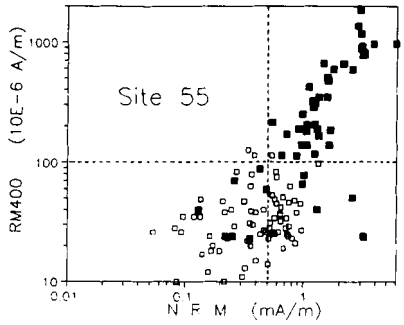


Figure 10. Remanence intenstiy after 400 °C thermal demagnetization (RM400) plotted versus the NRM intensity for site 55. See text for significance. Full squares: specimens showing a stable magnetite component (category 1, see text). Open squares: specimens showing no stable magnetite component (category 2, see text).

Table 2. Summary of the palaeomagnetic results of all sites shown in Fig. 2: declinations (Decl., °), inclinations (Incl., °), α_{95}(°) and precision parameters k for core and bedding corrected directions. Also NRM intensities and blocking temperatures (for thermally demagnetized sites) or peak fields (to obtain characteristic remanences for AF demagnetized sites) are listed. For the pre-folding remanence the north pole position (Plat., Plon., °) is calculated with error parameters dp(°) and dm(°). (*) predominantly magnetite, no/re: normal/reverse polarity, Nm: number of measured specimens, N: number of specimens included in Fisher statistics (generally one specimen from one core sample has been included).

A) PRE-FOLDING REMANENCE

SITE	N	(Nm)	POLARITY no	POLARITY re	BLOCK.TEMP. max (°C)	CORE CORRECTED DIRECTION Decl	Incl	alpha95	k	BEDDING CORRECTED DIRECTION Decl	Incl	alpha95	k	POLE POSITION Plat	Plon	dp	dm
55	40	(128)	40	-	500-550	338.9	-25.1	9.0	7.4	334.3	-54.1	4.2	30.2	22.4N	286.7E	4.1	5.9

B) POST-FOLDING REMANENCES (mainly normal polarity)

SITE	N	(Nm)	POLARITY no	POLARITY re	NRM (mA/m)	BLOCK.TEMP. max. (°C)	CORE CORRECTED DIRECTIONS Decl	Incl	alpha95	k	BEDDING CORRECTED DIRECTIONS Decl	Incl	alpha95	k
25	7	(11)	6	1	0.02-1.2	330-350	336.1	74.4	7.6	64.5	246.7	63.7	63.7	1.9
26	11	(11)	11	-	0.5-27	330-350	297.7	63.3	20.8	5.8	235.2	39.2	20.8	4.9
28	8	(8)	7	1	1.2-169	330-360	316.6	49.2	18.6	9.8	288.0	39.2	24.0	6.3
43	10	(12)	10	-	0.7-143	300-350	303.5	32.8	14.3	12.3	289.0	17.9	14.9	11.5
45	11	(12)	11	-	0.05-1.0	300-400	329.8	68.0	11.8	15.9	319.2	51.6	15.4	9.8
Mean	**5**	**Sites**					**312.9**	**58.3**	**17.6**	**19.8**	**277.4**	**46.3**	**28.8**	**8.0**

C) POST-FOLDING REMANENCES (mainly reversed polarity)

SITE	N	(Nm)	POLARITY no	POLARITY re	NRM (mA/m)	BLOCK.TEMP. max. (°C)	CORE CORRECTED DIRECTIONS Decl	Incl	alpha95	k	BEDDING CORRECTED DIRECTIONS Decl	Incl	alpha95	k
30	9	(10)	-	9	0.1-16	365-400	196.6	-59.2	10.4	25.5	130.2	-74.1	10.5	25.2
31	10	(10)	-	10	0.3-1.6	300-350	188.2	-72.9	7.2	46.0	142.5	-70.4	7.1	47.2
32	9	(10)	-	9	1.0-4.7	330-350	192.2	-59.0	7.0	55.6	178.4	-25.7	6.9	57.2
33	9	(9)	-	9	1.3-10	330-380	185.3	-67.3	3.8	179	194.3	-7.9	3.8	179
35	10	(10)	-	10	0.1-12	320-360	194.8	-63.6	9.1	29.0	93.7	-74.7	9.2	28.0
47	9	(10)	1	8	0.06-1.0	350-360	194.5	-67.3	8.0	42.0	192.3	-62.7	7.8	43.9
48	9	(10)	-	9	0.02-1.0	300-360	194.9	-72.6	5.3	94.9	187.9	-69.5	5.6	84.9
49	5	(10)	-	5	0.03-0.5	AF 20-50mT*	212.8	-63.5	10.8	50.8	198.8	-70.6	9.2	70.0
50	10	(10)	1	9	0.1-1.0	300-330	193.0	-61.1	10.6	21.6	199.1	-39.7	10.6	21.9
51	10	(10)	-	10	0.09-0.6	330-370	197.5	-70.0	3.7	174	191.5	-66.8	5.3	85.5
55	124	(128)	-	124	0.04-5.9	320-350	201.9	-60.7	1.8	51.3	191.7	-25.3	7.1	4.2
52	10	(10)	-	10	0.04-4.9	350-370	220.2	-68.6	6.1	62.8	227.7	-75.6	8.4	34.3
53	10	(10)	-	10	0.1-0.9	330-370	198.3	-66.6	5.4	80.9	87.5	-82.6	8.6	32.8
54	7	(9)	-	7	0.05-0.4	≈500*	191.3	-75.1	35.0	3.9	191.3	-75.1	35.0	3.9
59	13	(16)	-	13	0.4-32	AF 30-50mT*	178.4	-51.9	10.1	17.9	93.2	-67.8	10.2	17.4
Mean	**12**	**Sites**	**(exept**	**49,54,59)**			**196.4**	**-65.9**	**3.2**	**183**	**182.8**	**-60.2**	**16.4**	**8.0**

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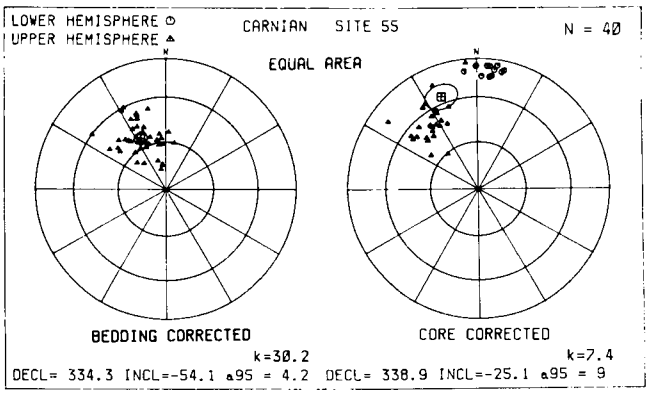


Figure 11. Equal area projection of the magnetite component (site 55) before and after bedding correction. A 99 per cent positive fold test is evident.

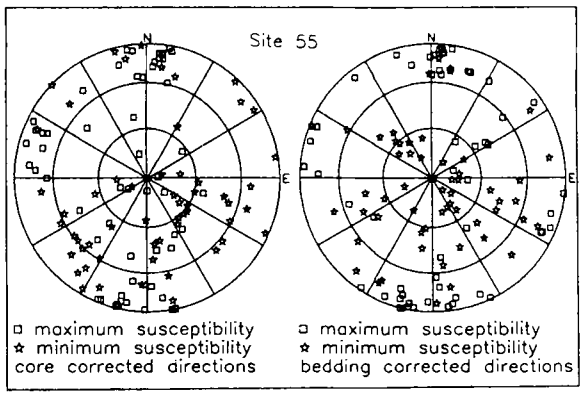


Figure 13. Anisotropy of susceptibility of site 55: equal area projection of the maximum and minimum susceptibility tensor of individual specimens before and after bedding correction. See test for significance.

AMS, shows only a weak alignment of minerals, and seems to have no relationships to the pyrrhotite or the magnetite directions.

INTERPRETATION

There is no obvious explanation for the origin and the tectonic significance of the aberrant post-folding secondary normal poling pyrrhotite component (cc: decl. = 312.9°, incl. = 58.3°). An interpretation is not attempted in this paper, but a local effect might account for it.

The magnetite component (bc: decl. = 334.3°, incl. = −54.1°) pre-dates the Himalayan folding and is probably of primary origin (detrital remanence of Lower Carnian age). In Table 3 expected and observed palaeolatitudes are compared. The observed palaeolatitude of site 55 corresponds to a position of the Manang area at the northern margin of India in the Lower Carnian, when India was still part of Gondwana. From pole positions for stable India, a palaeoinclination of 41.5°S is expected for the Manang area on the basis of no movement relative to stable India. Our result also coincides with Triassic palaeolatitudes from the TSS of the adjacent Thakkhola area (Klootwijk & Bingham 1980). Also, observed inclinations from the Triassic TSS of Kashmir (Klootwijk *et al.* 1983) agree with expected values for the Early Triassic, but a non-explicable 30.5° difference exists for the Late Triassic. The extent of the Triassic northern India margin cannot be reconstructed reliably from the inclination data. The strong anticlockwise rotation of

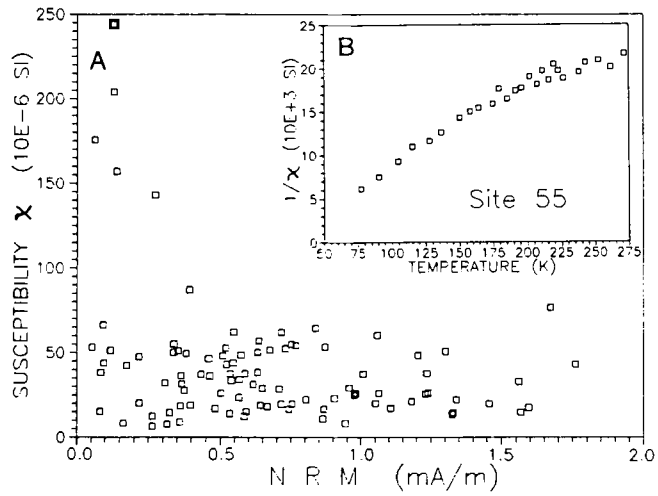


Figure 12. Susceptibility χ plotted versus the NRM (A), and $1/\chi$ plotted versus temperature (B) for site 55. See text for significance.

Table 3. Observed (obs.) and expected (exp.: calculated from the Indian APWP) declinations(°) and inclinations(°) for Central Nepal and Kashmir for 20–30 Ma (mean), 40–50 Ma (mean), Triassic Tr1–Tr3 (values rounded). (1) This paper. (2) Calculated from Indian Ocean DSDP cores and from sea floor spreading (SFS: Indian plate movement relative to a fixed hotspot frame) (Klootwijk 1979; Duncan 1981; Morgan 1983; Patriat & Achache 1984; data summarized by Klootwijk *et al.* 1985). Errors of SFS data are estimated by Besse *et al.* (1984) to be of the order of 3°. (3) Calculated from Pachmarhi Ss (Wensink 1968) and Parsora Ss (Bhalla & Verma 1969). (4) Calculated from Panchet Clay Beds (Klootwijk 1974), Mangli Red Beds (Wensink 1968), Wardha Valley Ss (Klootwijk 1975) and Kamthi Red Beds (Wensink 1968). (5) Thakkhola (Klootwijk 1980). (6) Kashmir Valley (Klootwijk *et al.* 1983).

DECLINATIONS

Age	North-Pole Position Plat Plon	Central Nepal (28.5N,84.0E) Exp.Decl	Obs.Decl	d(Decl)
20-30ma	78N(2)305E	352	17(1)	25E
40-50ma	62N(2)303E	343	17(1)	34E
Tr_3	20N(3)311E	312	335(1)	23E

INCLINATIONS

Age	North-Pole Position Plat Plon	Central Nepal (28.5N,84.0E) Exp.Incl	Obs.Incl	Kashmir (30.0N,75.0E) Exp.Incl	Obs.Incl
Tr_3	20N(3)311E	-42	-54(1) -48(5) -43(5) -47(5)	-28	-58(6)
Tr_2					-58(6)
P_2-Tr_1	10S(4)306E			-58	-69(6) -64(6)

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stable India (48°) since the Triassic reduces the meaningfulness of palaeolatitudes and increases the importance of palaeolongitudes.

The steep cc inclination of the reverse pyrrhotite component (cc: decl. = 196.4°, incl. = −65.9°) can be interpreted in terms of a tectonic tilt of the TSS in at least the Manang complex. The inclination of the present dipole field at the Manang area is 47°. Due to the drift history of India the maximum northern position of the TSS is the present one. Accordingly, the inclination of (−)66° infers that a minimum 19° northward tilt of the TSS and probably of the whole Himalayan crystalline crust up to the MCT (surfacing 20–30 km south of the investigated area) has occurred after remanence acquisition (Fig. 14). The degree of the tilt agrees approximately with the 20°–30° dip of the MCT suggested by focal depths of earthquakes (Seeber, Armbruster & Qittmeyer 1981; Teng Jiwen *et al.* 1987). A higher degree of tilt has to be invoked if remanence acquisition occurred in a more southerly latitude than the present one. Very recent tilt of the TSS is suggested from the Quaternary relief formation (Jaros & Kalvoda 1978). Note that the proposed 19° tilt is based on the assumption that the rotation plane of the reverse pyrrhotite direction was perpendicular to the MCT which is expected to be parallel to the rotation axis. In this case the declination was not changed by the tilt.

It is not evident from our data whether the tilt represents a local or a regional effect. However Klootwijk & Bingham (1980) and Rochette (1990) also report inclination steepening from the TSS of the Thakkhola area (6°) and from the Everest leucogranite (10°) respectively. Hence a regional northward tilt of the tectonic unit north of the MCT is suggested, but more data are needed to confirm the regional significance.

Klootwijk *et al.* (1985) report clockwise deviations of observed declinations with respect to expected values in the central Himalaya (Thakkhola, Garwhal) and especially in the western Himalaya. Expected declinations are calculated from the Indian APWP on the supposition of no relative movement of the areas relative to stable India. Klootwink *et al.* interpret this deviation in terms of rotational underthrusting of stable India under the Tibetan Plateau

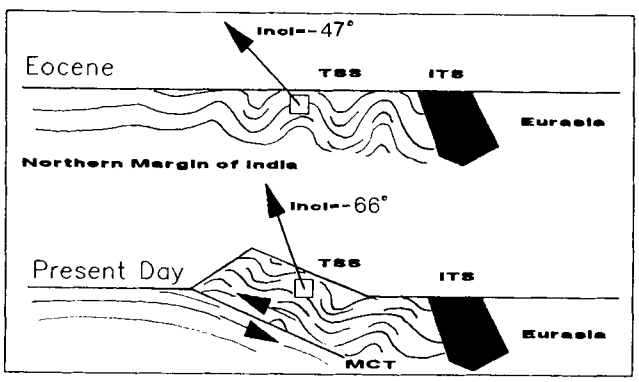


Figure 14. Inclination steepening of the reverse pyrrhotite component by tilt of the Tibetan Sedimentary Series (TSS) along the Main Central Thrust (MCT). The pyrrhotite component was acquired after folding and before tilt at the MCT. See text for discussion.

along the MCT (by rotation of the TSS around a pivot at 35°N/73°E) and calculated a magnitude of underthrusting at the longitude of the Thakkhola (84°E) area of 650 km.

The declination of the pre-folding magnetite from site 55 deviates clockwise from the expected value by 23° (Table 3). However the measured declination has not been corrected for the plunge of the fold axis (not possible to identify). Hence the declination of the post-folding pyrrhotite remanence seems to be the more suitable one to estimate the magnitude of rotational underthrusting in the Manang area. Dealing with the secondary pyrrhotite declination requires knowledge of the remanence age in order to find the reference (expected) declination, Klootwink *et al.* (1985) noticed that declination data of remanence directions (secondary and primary ones) within the Himalayan zone show a consistent pattern (following oroclinal bending) and concluded that a regional tectonic significance of the declination data is likely. The following discussion on the remanence acquisition age and the magnitude of rotational underthrusting relies on this regional tectonic significance and neglects local effects.

Folding of the TSS was certainly active up to slightly younger ages than the India–Eurasia collision (about 40–50 Ma). Due to the post-folding age 45 Ma can be regarded as an upper limit for the remanence age of reverse pyrrhotite component.

The declination of the reverse pyrrhotite component suggests a 16.4° clockwise rotation of the Manang area since the time of remanence acquisition. This rotation is opposite to that of India and has probably been caused by the clockwise rotation of the Eurasian plate (EP) after the continental contact of the TSS and the EP. This 16.4° rotation corresponds to the degree of the EP rotation within the last 50 Ma (Eurasian APWP of Irving & Irving 1982). Consequently a remanence age of 50 Ma is concluded. From considerations of error limits for rotations of the EP and the Manang area a remanence age of 30 Ma is still significant but a 20 Ma age is not significant at the 95 per cent confidence level.

In summary, the age of remanence acquisition of the reverse pyrrhotite component is suggested to be between 45 and 25 Ma. Further evidence for a lower limit of 25 Ma results from a cooling event at the Oligocene–Miocene boundary (Le Fort 1975). Blocking of the pyrrhotite component can be related to this cooling. The onset of underthrusting at the MCT is likely to be the cause of the cooling and possibly may indicate a change from folding to thrusting.

The declination of the reverse pyrrhotite component deviates from the expected declination by 25° and 34° for a 25 and a 45 Ma remanence age respectively (Table 3). Besides the remanence age, the magnitude of underthrusting at the longitude of central Nepal (CN) depends on whether oroclinal bending occurred before or after underthrusting started in this area (Fig. 15).

Oroclinal bending is likely to have occured during the initial continental collision by a wedge like indentation of NW India into Eurasia, whereas subduction of oceanic crust continued in the east for some time. Hence the shape of the mountain arc would have already existed at the time when rotational underthrusting of continental crust started in CN (Fig. 15, stage B). This is, in principal, identical to

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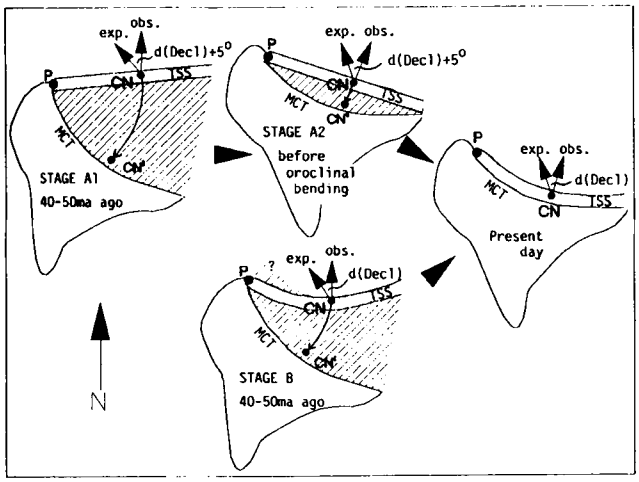


Figure 15. Underthrusting of the former northern margin of India following Klootwijk *et al.*'s (1985) rotational underthrusting and oroclinal bending model. Two alternative developments are shown, dependent on whether underthrusting at central Nepal (CN) post-dates (stage B to present) or pre-dates (stages A1, A2 to present) oroclinal bending. Observed (obs.) palaeodeclinations (remanence acquisition at the true position CN for stages A1 or B) and expected (exp.) palaeodeclinations (remanence acquisition at the present relative position to stable India CN' at the stages A1 or B) are drawn. The hatched area represents the underthrusted former northern leading edge of India (see text for further explanations). CN: actual position of Central Nepal for the different tectonic stages; CN': present position of Central Nepal relative to stable India, P: pivot, d(Decl.): angle between observed and expected declinations; TSS: Tibetan Sedimentary Series, MCT: Main Central Thrust.

Klootwijk *et al.*'s (1985) model. In this case the magnitude of rotational underthrusting at CN is then given by the arc of circle from CN to CN' and is calculated from $2\pi rd(\text{Decl.})°/360°$ [$r = 1200$ km: distance from the pivot to CN; d(Decl.) see Table 3]. For remanence ages of 45 Ma and 25 Ma magnitudes of 710 ± 150 km and 520 ± 150 km respectively result for CN. Declination errors are deduced from α_{95} values with inclinations taken into account. The remanence age must pre-date the onset of the oroclinal bending formation to explain the observed declination pattern along the Himalayan arc (Klootwink *et al.* 1985). This favours a 45 Ma age but a 25 Ma age cannot be excluded. The declination pattern is mainly characterized by strong clockwise rotations in the western Himalaya, but the remanence ages in this part must not necessarily coincide with the remanence age of the secondary reverse pyrrhotite component at CN.

It is less likely that the formation of oroclinal bending started after completion of rotational underthrusting. In this case the distance CN–CN' (about 300 km) has to be added to the magnitude of rotational underthrusting (stage A2 of Fig. 15). The true angle of declination deviation could possibly be increased to about d(Decl.) $\pm 5°$ by keeping the angle between the MCT and the observed declination fixed and rebending the Himalayan arc to a straight line. For a remanence age of 45 and 25 Ma the magnitude of underthrusting by rotational underthrusting plus oroclinal bending will be about 1100 ± 200 km and 900 ± 200 km respectively.

Of course, any higher magnitude of continental underthrusting is possible by straight forward subduction. Hence the final significant result is a minimum magnitude of 520 ± 150 km continental underthrusting at the longitude of central Nepal (84°E). The accuracy of the result is quite unsatisfactory but could be considerably improved if there are convincing arguments of a more precise dating of the remanence age and for the tectonic mode of underthrusting and oroclinal bending. A pure validity of either model A or B is not likely. Probably the formation of oroclinal bending started with the initial collision but progressing developed to the present shape during the entire period of continental crustal shortening.

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